Measure Theoretic Probability
Lesson 16
Random Vectors and $\sigma(X_1, \ldots, X_n)$

Kirschner Awonatey Amaanya

Department of Statistics and Actuarial Science, KNUST

Outline

1. Simple Random Vectors
2. The Main Theorem
3. Proof of Part (I)
4. Proof of Part (II)
5. Summary and Exercises

Outline

1. Simple Random Vectors
2. The Main Theorem
3. Proof of Part (I)
4. Proof of Part (II)
5. Summary and Exercises

Two-Dimensional Simple Random Vectors

Setting

$(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space. We work with **two** components for clarity; n components behave identically. We stay in the **simple** setting.

Definition

$\mathrm{X} = (X_1, X_2) : \Omega \to \mathbb{R}^2$, $\mathrm{X}(\omega) = \big(X_1(\omega), X_2(\omega)\big)$, is a **simple random vector** if

1. it takes only **finitely many** values in $\mathbb{R}^2$, and
2. for every point $(x_1, x_2) \in \mathbb{R}^2$,

$$\{\omega : \ X_1(\omega) = x_1, \ X_2(\omega) = x_2\} = \{\mathrm{X} = (x_1, x_2)\} \in \mathcal{F}.$$

Remark

Condition (2) is the measurability condition: point pullbacks must be measurable, exactly as for a single simple random variable.

Two Definitions of $\sigma(X_1, X_2)$

Definition A (pull back vector points)

$\sigma\big((X_1, X_2)\big)$ is the **smallest** σ-field on Ω for which $\mathrm{X} = (X_1, X_2)$ is measurable: all sets $\{\mathrm{X} = (x_1, x_2)\}$, $(x_1, x_2) \in \mathbb{R}^2$, lie in it.

Definition B (pull back each component)

$\sigma(X_1, X_2)$ is the **smallest** σ-field for which **each** X_1 and X_2 is individually measurable: all sets $\{X_1 = x_1\}$ and $\{X_2 = x_2\}$ lie in it.

Notation

Double parentheses $\sigma((X_1, X_2))$ mean A; single parentheses $\sigma(X_1, X_2)$ mean B. Many authors ignore the difference. The next frames show that they are the **same** σ-field.

The Two Definitions Agree

$(B \Rightarrow A)$ Vector points are in the σ-field of Definition B

Let $\mathcal{S} = \sigma(X_1, X_2)$ be as in B and fix (x_1, x_2). Then

$$\{\mathrm{X} = (x_1, x_2)\} = \{X_1 = x_1\} \cap \{X_2 = x_2\} \in \mathcal{S},$$

an intersection of two sets in a σ-field. So X is $\mathcal{S}$-measurable, and by minimality $\sigma((X_1, X_2)) \subseteq \mathcal{S}$.

$(A \Rightarrow B)$ Components are in the σ-field of Definition A

Let $\mathcal{V} = \sigma((X_1, X_2))$ be as in A and fix x_1. Letting x_2 range over the **finitely many** values taken by X_2,

$$\{X_1 = x_1\} = \bigcup_{x_2} \{\mathrm{X} = (x_1, x_2)\} \in \mathcal{V},$$

a finite union of sets in $\mathcal{V}$. Likewise for X_2. So both components are $\mathcal{V}$-measurable, and by minimality $\sigma(X_1, X_2) \subseteq \mathcal{V}$.

Conclusion

$\sigma((X_1, X_2)) = \sigma(X_1, X_2)$. The union is finite precisely because the variables are simple.

Outline

1. Simple Random Vectors
2. The Main Theorem
3. Proof of Part (I)
4. Proof of Part (II)
5. Summary and Exercises

Statement (Vector Version)

Theorem

Let $X_1, \ldots, X_n$ be simple random variables on $(\Omega, \mathcal{F}, \mathbb{P})$.

(I) $\sigma(X_1, \ldots, X_n)$ consists of all sets

$$\{(X_1, \ldots, X_n) \in H\} = \{\omega : \ (X_1(\omega), \ldots, X_n(\omega)) \in H\}, \qquad H \subseteq \mathbb{R}^n \text{ arbitrary.}$$

(II) A simple random variable Y is $\sigma(X_1, \ldots, X_n)$-measurable if and only if

$$Y = f(X_1, \ldots, X_n) \qquad \text{for some function } f : \mathbb{R}^n \to \mathbb{R}.$$

Remarks

- Taking $n = 1$ gives the theorem stated in Lesson 15.
- **Shorthand:** we often suppress the ω's and write $\{(X_1, \ldots, X_n) \in H\}$ for the set above.
- Since the variables are simple, it is enough to pull back finite subsets H of $\mathbb{R}^n$.

Outline

1. Simple Random Vectors
2. The Main Theorem
3. Proof of Part (I)
4. Proof of Part (II)
5. Summary and Exercises

Proof of (I): Strategy

Define the candidate class

$$\mathcal{M} = \{\{(X_1, \ldots, X_n) \in H\} : \ H \subseteq \mathbb{R}^n\}.$$

We must show $\mathcal{M} = \sigma(X_1, \ldots, X_n)$.

Three steps

1. $\mathcal{M}$ is a σ-field.
2. Each X_i is $\mathcal{M}$-measurable.
3. $\mathcal{M} \subseteq \sigma(X_1, \ldots, X_n)$.

Why this suffices

By 1 and 2, $\mathcal{M}$ is a σ-field making every X_i measurable, so by minimality $\sigma(X_1, \ldots, X_n) \subseteq \mathcal{M}$. Step 3 gives the reverse inclusion. Hence equality.

Step 1: $\mathcal{M}$ Is a σ-Field

Contains Ω

Take $H = \mathbb{R}^n$. Every ω is mapped into $\mathbb{R}^n$, so $\{(X_1, \ldots, X_n) \in \mathbb{R}^n\} = \Omega$.

Closed under complements

Let $A = \{(X_1, \ldots, X_n) \in H\} \in \mathcal{M}$. Each $\omega \in A$ maps into H and each $\omega \notin A$ maps outside H, so

$$A^c = \{(X_1, \ldots, X_n) \in H^c\} \in \mathcal{M}.$$

Closed under countable unions

Let $A_j = \{(X_1, \ldots, X_n) \in H_j\} \in \mathcal{M}$ for $j = 1, 2, \ldots$. A point maps into some H_j exactly when it maps into $\bigcup_j H_j$, so

$$\bigcup_j A_j = \left\{(X_1, \ldots, X_n) \in \bigcup_j H_j\right\} \in \mathcal{M}. \qquad \square$$

Steps 2 and 3

Step 2: each X_i is $\mathcal{M}$-measurable

X_i is simple, so it suffices to pull back single points. For $x \in \mathbb{R}$ let

$$H = \{(y_1, \ldots, y_n) \in \mathbb{R}^n : \ y_i = x\}.$$

Then $\{X_i = x\} = \{(X_1, \ldots, X_n) \in H\} \in \mathcal{M}$. □

Step 3: $\mathcal{M} \subseteq \sigma(X_1, \ldots, X_n)$

Let $A = \{(X_1, \ldots, X_n) \in H\} \in \mathcal{M}$. Only finitely many points of H are actually attained by the vector, so

$$A = \bigcup_{x \in H \cap \text{range}} \{(X_1, \ldots, X_n) = x\},$$

a **finite** union. Each set $\{(X_1, \ldots, X_n) = x\}$ lies in $\sigma(X_1, \ldots, X_n)$ (Definition A, equivalent to B by the earlier frame, extended to n components), and a σ-field is closed under finite unions. Hence $A \in \sigma(X_1, \ldots, X_n)$. □

Conclusion

$\mathcal{M} = \sigma(X_1, \ldots, X_n)$, which proves (I).

Outline

1. Simple Random Vectors
2. The Main Theorem
3. Proof of Part (I)
4. **Proof of Part (II)**
5. Summary and Exercises

($\Rightarrow$) Measurable $\Longrightarrow$ a Function of the X_i

Setup

Let Y be simple and $\sigma(X_1, \ldots, X_n)$-measurable, with distinct values $a_1, \ldots, a_k$. Then $\{Y = a_i\} \in \sigma(X_1, \ldots, X_n)$.

Use part (I)

For each i there is $H_i \subseteq \mathbb{R}^n$ with

$$\{Y = a_i\} = \{(X_1, \ldots, X_n) \in H_i\}.$$

Replace H_i by $H_i' = H_i \cap \mathrm{range}(X_1, \ldots, X_n)$; the set above does not change. The H_i' are **pairwise disjoint**: a point in $H_i' \cap H_j'$ would force $Y = a_i$ and $Y = a_j$ at some ω, impossible for $a_i \neq a_j$.

Define f

$$f(\mathrm{x}) = \sum_{i=1}^{k} a_i \, I_{H_i'}(\mathrm{x}), \qquad \mathrm{x} \in \mathbb{R}^n.$$

If $\omega \in \{Y = a_i\}$, then $(X_1(\omega), \ldots, X_n(\omega)) \in H_i'$, so $f(X_1(\omega), \ldots, X_n(\omega)) = a_i = Y(\omega)$. Since the sets $\{Y = a_i\}$ cover Ω, $Y = f(X_1, \ldots, X_n)$. □

($\Leftarrow$) A Function of the $X_i \Longrightarrow$ Measurable

Setup

Suppose $Y = f(X_1, \ldots, X_n)$ for some $f : \mathbb{R}^n \to \mathbb{R}$ (and Y is simple, since $(X_1, \ldots, X_n)$ takes finitely many values). We must show $\{Y = y\} \in \sigma(X_1, \ldots, X_n)$ for every $y \in \mathbb{R}$.

Proof

Fix $y \in \mathbb{R}$ and let $H = \{\mathrm{x} \in \mathbb{R}^n : \ f(\mathrm{x}) = y\}$. Then

$$\{Y = y\} = \{\omega : \ f(X_1(\omega), \ldots, X_n(\omega)) = y\} = \{(X_1, \ldots, X_n) \in H\},$$

which is in $\sigma(X_1, \ldots, X_n)$ by part (I). □

Interpretation

Y is $\sigma(X_1, \ldots, X_n)$-measurable exactly when Y is determined by the values of $X_1, \ldots, X_n$: knowing the X_i lets you compute Y.

Outline

1. Simple Random Vectors
2. The Main Theorem
3. Proof of Part (I)
4. Proof of Part (II)
5. Summary and Exercises

Summary

Key results

Concept / Result	Content
Simple random vector	finitely many values, $\{\mathrm{X} = \mathrm{x}\} \in \mathcal{F}$ for all $\mathrm{x} \in \mathbb{R}^n$
Two definitions	pulling back vector points $\iff$ pulling back components (finite unions/intersections)
Theorem (I)	$\sigma(X_1, \ldots, X_n) = \left\{\{(X_1, \ldots, X_n) \in H\} : \ H \subseteq \mathbb{R}^n\right\}$
Proof of (I)	$\mathcal{M}$ is a σ-field, each X_i is $\mathcal{M}$-measurable, $\mathcal{M} \subseteq \sigma$
Theorem (II)	Y is $\sigma(X_1, \ldots, X_n)$-measurable $\iff$ $Y = f(X_1, \ldots, X_n)$
Notation	suppress the ω's: $\{(X_1, \ldots, X_n) \in H\}$

Exercises

1. Let Ω be the eight outcomes of three coin flips, $X_1 =$ indicator that flip 1 is heads, $X_2 =$ indicator that flip 2 is heads. List the sets $\{(X_1, X_2) = (x_1, x_2)\}$ and describe $\sigma(X_1, X_2)$.
2. In Exercise 1, let Y be the number of heads among the first two flips. Write $Y = f(X_1, X_2)$ explicitly.
3. Is $Z = I_{\{\text{flip 3 is heads}\}}$ a function of (X_1, X_2)? Is it $\sigma(X_1, X_2)$-measurable?
4. Show that $\sigma(X_1) \subseteq \sigma(X_1, X_2)$ and $\sigma(X_2) \subseteq \sigma(X_1, X_2)$.
5. Show that $\sigma(X_1, \ldots, X_n) = \sigma\big(\sigma(X_1) \cup \cdots \cup \sigma(X_n)\big)$.
6. In the proof of $(\Rightarrow)$ in part (II), why must we intersect H_i with the range of the vector before defining f?

Outlook

Next lesson

A relationship between $\sigma(X_1, X_2)$ and the individual fields $\sigma(X_1)$, $\sigma(X_2)$. Then we leave the simple case and treat **general** random variables, and afterwards **convergence** of sequences of random variables.

Looking ahead

σ-fields generated by several random variables are the language of independence of random variables and, later, of the tail σ-field and Kolmogorov's 0–1 law for random variables.